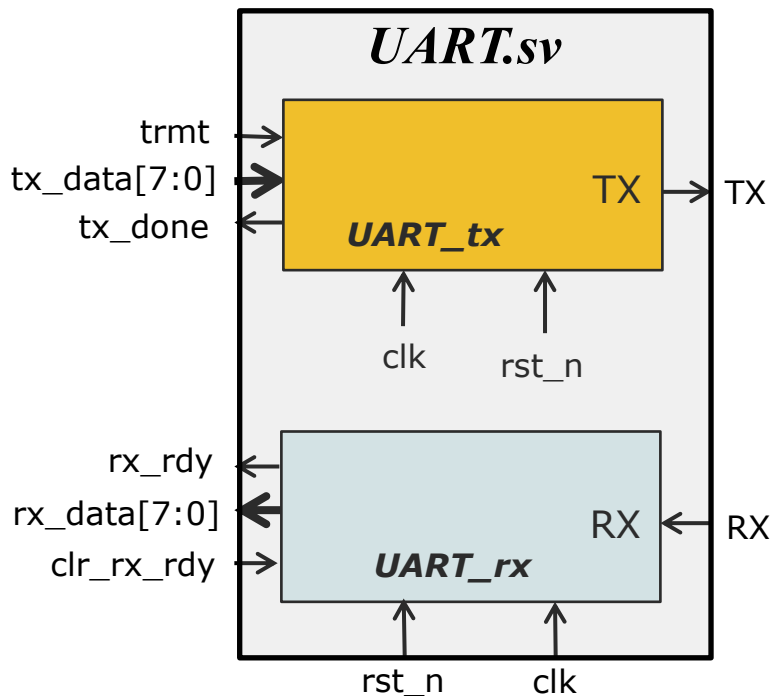




Exercise 16: Synthesizing UART

- From Ex14, you should have a UART transceiver (***UART.sv***) that makes use of your ***UART_tx*** and ***UART_rx*** blocks
 - We will synthesize this block using Synopsys Design Compiler targeting a standard-cell library
- You may want to go through the Synopsys Tutorial first



Signal	Dir	Description
clk,rst_n	in	50MHz clk, active low asynch reset
trmt	in	Initiates UART transmissison
tx_data[7:0]	in	Data to transmit
tx_done	out	Indicates transmission complete
rx_rdy	out	New 8-bit data ready
clr_rx_rdy	in	Knock down rdy flag
rx_data[7:0]	out	8-bit data received
TX	out	Serial data out line
RX	in	Serial data in line



Exercise 16: Synthesizing UART

- Write a synthesis script (**UART.dc**) to synthesize your UART. The script should perform the following:
 - Defines a clock of 500MHz frequency and sources it to clock
 - Performs a set don't touch on the clock network
 - Defines input delays of 0.4 ns on all inputs other than clock
 - Defines a drive strength equivalent to a 2-input nand of size 2 from the Synopsys 32nm library (NAND2X2_LVT) for all inputs except clock and rst_n
 - Defines an output delay of 0.4ns on all outputs.
 - Defines a 0.10pf load on all outputs.
 - Sets a max transition time of 0.15ns on all nodes.
 - Employs the Synopsys 32nm wire load model for a block of size 16000 sq microns
 - Compiles, then flattens the design so it has no hierarchy, and compiles again.
 - Produces a min_delay & max delay report
 - Produces an area report
 - Writes out the gate level verilog netlist (**UART.vg**)
- **Submit:**
 - Your synthesis scripts (**UART.dc**)
 - The output reports for area (**UART_area.txt**)
 - The gate level verilog netlist (**UART.vg**)

Your cell area should be in the ballpark of 720 and total area 900 (+/- 150)



Additional Instructions

- Synopsys Design Compiler is only on the CAE Linux machines
 - Remote into them using SSH (e.g., use clients like PuTTY or MobaXterm)
 - Under the “Tutorials” section of the Canvas page there is a video about accessing CAE linux machines remotely. There is another one about FTPing files.

Steps:

- SSH into: **best-tux.cae.wisc.edu**
- Make a directory with your source files:
 - mkdir ece551/exercise16
 - Copy UART.sv, UART_rx.sv, UART_tx.sv into folder
- Kick off Synopsys Design compiler in command shell mode: **dc_shell**
- Type/paste each line of the script into dc_shell command line and make sure there are no errors
(once you’ve verified no errors, you can just source the script for future runs)

Here showing how to login using PuTTY

